

2026 APMCM Control Sheet

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2026 APMCM summary sheet

[Summary: problem overview + methods for each sub-problem + key numerical results + conclusions, 300–500 words]

Keywords: Keyword1; Keyword2; Keyword3

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I. Introduction

1.1 Problem Background

With the continuous advancement of global industrialization and population growth, energy consumption has been steadily increasing. Although the proportion of clean energy from solar, wind, and nuclear power has risen, fossil fuels remain the dominant energy source, leading to the exacerbation of the urban heat island effect and greenhouse effect.

Passive Daytime Radiative Cooling (PDRC) has emerged as a transformative zero-energy cooling technology. Based on the Stefan-Boltzmann Law and the 8–13 μm atmospheric transparency window, PDRC enables objects to spontaneously emit thermal radiation into outer space (approximately 3 K), achieving sub-ambient cooling without any energy input. Among candidate materials, polydimethylsiloxane (PDMS) thin films exhibit ultra-high visible transmittance and high infrared emissivity arising from Si–O–Si and Si–C bond vibrations, making them highly suitable for radiative cooling applications in energy-efficient buildings, solar cells, and personal thermal management.

However, the radiative cooling performance depends critically on film thickness and structural design, governed by thin-film optical interference effects. This paper establishes a comprehensive modeling framework based on the Transfer Matrix Method (TMM) and Kirchhoff's law, extending from single-layer emissivity analysis to multi-layer structure optimization with economic feasibility assessment.

1.2 Problem Restatement

This paper addresses four interconnected sub-problems:

Problem 1: Establish a mathematical model to study the emissivity of PDMS thin films as a function of wavelength for different film thicknesses.

Problem 2: Develop an evaluation model to assess the radiative cooling performance of PDMS thin films with varying thicknesses, providing recommendations for technology advancement.

Problem 3: Select suitable materials to form multilayer film structures with PDMS, incorporating optimization design algorithms for optimal material selection and layer thicknesses.

Problem 4: Perform comprehensive optimal design of radiative cooling materials and structures to achieve the best possible cooling performance, followed by feasibility and cost evaluation.

1.3 Overall Approach

Our approach proceeds through four stages: (1) single-layer PDMS emissivity modeling via TMM with Airy summation; (2) radiative cooling performance evalua-

tion incorporating blackbody radiation, atmospheric transmittance, and solar spectrum; (3) multilayer structure design with parameter scanning across candidate materials; and (4) comprehensive optimization with economic and feasibility assessment.

II. Model Assumptions

To simplify the problem while preserving physical accuracy, we make the following assumptions:

1. **Smooth planar interfaces:** All film interfaces are assumed to be perfectly smooth and parallel. Surface roughness effects on optical scattering are neglected, as RMS roughness is typically much smaller than the relevant wavelengths.
2. **Isotropic and homogeneous media:** Each layer material is assumed to be optically isotropic and homogeneous. The complex refractive index $\tilde{n}(\lambda) = n(\lambda) + ik(\lambda)$ fully characterizes the optical response at each wavelength.
3. **Coherent thin-film interference:** For film thicknesses below the coherence length (100 μm), optical interference effects are treated coherently using the Airy summation formula. For very thick layers, the incoherent limit is approached.
4. **Normal and near-normal incidence approximation:** For radiative transfer calculations, we integrate over the hemisphere using Gauss-Legendre quadrature to account for angular dependence of emissivity.
5. **Kirchhoff's law applicability:** We invoke Kirchhoff's law of thermal radiation, equating spectral directional absorptivity to spectral directional emissivity: $\varepsilon(\lambda, \theta) = A(\lambda, \theta) = 1 - R(\lambda, \theta) - T(\lambda, \theta)$.
6. **Steady-state thermal conditions:** The radiative cooler is assumed to operate under steady-state thermal equilibrium, with net cooling power $P_{\text{cool}} = 0$ determining the equilibrium temperature.
7. **Standard atmospheric and solar conditions:** Atmospheric transmittance follows the MODTRAN mid-latitude summer model. Solar irradiance follows the ASTM G173-03 AM1.5 Global Tilt spectrum (1000 W/m^2 total).
8. **Non-radiative heat transfer:** Conduction and convection are modeled as a lumped linear term $P_{\text{nonrad}} = h_c(T_{\text{amb}} - T)$, with $h_c = 6.9 \text{ W}/(\text{m}^2 \cdot \text{K})$ for natural convection.

表 1: Key Notation

Symbol	Description	Unit
λ	Wavelength	μm
θ	Incidence angle from normal	rad
$\tilde{n} = n + ik$	Complex refractive index	—
d	Film thickness	μm
$R(\lambda, \theta)$	Spectral reflectance	—
$T(\lambda, \theta)$	Spectral transmittance	—
$\varepsilon(\lambda, \theta)$	Spectral directional emissivity	—
$\bar{\varepsilon}_{8-13}$	Average emissivity in atmospheric window	—
$I_{BB}(\lambda, T)$	Planck blackbody spectral radiance	$\text{W}/(\text{m}^2 \cdot \mu\text{m} \cdot \text{sr})$
P_{rad}	Outgoing thermal radiation power	W/m^2
P_{atm}	Absorbed atmospheric radiation	W/m^2
P_{sun}	Absorbed solar radiation	W/m^2
P_{nonrad}	Non-radiative heat transfer	W/m^2
P_{cool}	Net radiative cooling power	W/m^2
T_{amb}	Ambient temperature (300 K)	K
ΔT	Temperature drop below ambient	K

III. Model Design and Methodology

3.1 Notation

3.2 Problem 1: Single-Layer PDMS Emissivity Model

3.2.1 Transfer Matrix Method

For a single thin film with complex refractive index $\tilde{n} = n + ik$ and thickness d between semi-infinite incident medium (air, $n_0 = 1$) and exit medium (n_2), the optical response is computed using the Airy summation formula:

$$r = \frac{r_{01} + r_{12} e^{2i\beta}}{1 + r_{01}r_{12} e^{2i\beta}}, \quad t = \frac{t_{01}t_{12} e^{i\beta}}{1 + r_{01}r_{12} e^{2i\beta}} \quad (1)$$

where the phase factor is:

$$\beta = \frac{2\pi}{\lambda} \tilde{n} d \cos \theta \quad (2)$$

The Fresnel reflection and transmission coefficients at each interface, for s - and p -polarizations, are:

$$r_{mn}^{(s)} = \frac{\tilde{n}_m \cos \theta_m - \tilde{n}_n \cos \theta_n}{\tilde{n}_m \cos \theta_m + \tilde{n}_n \cos \theta_n}, \quad r_{mn}^{(p)} = \frac{\tilde{n}_n \cos \theta_m - \tilde{n}_m \cos \theta_n}{\tilde{n}_n \cos \theta_m + \tilde{n}_m \cos \theta_n} \quad (3)$$

$$t_{mn}^{(s)} = \frac{2\tilde{n}_m \cos \theta_m}{\tilde{n}_m \cos \theta_m + \tilde{n}_n \cos \theta_n}, \quad t_{mn}^{(p)} = \frac{2\tilde{n}_m \cos \theta_m}{\tilde{n}_n \cos \theta_m + \tilde{n}_m \cos \theta_n} \quad (4)$$

Angles are related through Snell's law: $\tilde{n}_m \sin \theta_m = \tilde{n}_n \sin \theta_n$.

The reflectance and transmittance for unpolarized light are:

$$R = \frac{1}{2} (|r^{(s)}|^2 + |r^{(p)}|^2), \quad T = \frac{1}{2} (T^{(s)} + T^{(p)}) \quad (5)$$

By Kirchhoff's law, the spectral directional emissivity equals absorptivity:

$$\varepsilon(\lambda, \theta) = A(\lambda, \theta) = 1 - R(\lambda, \theta) - T(\lambda, \theta) \quad (6)$$

For an opaque substrate ($T = 0$), this simplifies to $\varepsilon = 1 - R$.

3.2.2 PDMS Optical Constants

The complex refractive index of PDMS is modeled using a multi-oscillator Drude-Lorentz dielectric function:

$$\varepsilon(\omega) = \varepsilon_\infty + \sum_{j=1}^M \frac{\omega_{p,j}^2}{\omega_{0,j}^2 - \omega^2 - i\gamma_j\omega} \quad (7)$$

where $\tilde{n} = \sqrt{\varepsilon}$ and $\omega = 2\pi c/\lambda$. The oscillator parameters capture the key Si–O–Si asymmetric stretching (9.26 μm), CH₃ symmetric deformation (7.94 μm), and Si–C rocking (12.5 μm) vibrational modes. In the solar spectral range (0.3–5 μm), PDMS is essentially transparent with $k < 10^{-5}$.

3.3 Problem 2: Radiative Cooling Performance Model

3.3.1 Net Cooling Power

The net radiative cooling power is the balance of four components:

$$P_{\text{cool}}(T) = P_{\text{rad}}(T) - P_{\text{atm}}(T_{\text{amb}}) - P_{\text{sun}} - P_{\text{nonrad}}(T) \quad (8)$$

3.3.2 Outgoing Thermal Radiation

$$P_{\text{rad}}(T) = \int_0^\infty d\lambda \int_0^{\pi/2} 2\pi \sin \theta \cos \theta I_{BB}(\lambda, T) \varepsilon(\lambda, \theta) d\theta \quad (9)$$

The Planck blackbody spectral radiance is:

$$I_{BB}(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_B T)} - 1} \quad (10)$$

where $h = 6.626 \times 10^{-34}$ J·s, $c = 2.998 \times 10^8$ m/s, and $k_B = 1.381 \times 10^{-23}$ J/K.

3.3.3 Atmospheric Counter-Radiation

$$P_{\text{atm}}(T_{\text{amb}}) = \int_0^\infty d\lambda \int_0^{\pi/2} 2\pi \sin \theta \cos \theta I_{BB}(\lambda, T_{\text{amb}}) \varepsilon(\lambda, \theta) \varepsilon_{\text{atm}}(\lambda, \theta) d\theta \quad (11)$$

The atmospheric emissivity is modeled from MODTRAN transmittance:

$$\varepsilon_{\text{atm}}(\lambda, \theta) = 1 - t_{\text{atm}}(\lambda)^{1/\cos \theta} \quad (12)$$

3.3.4 Solar Absorption and Non-Radiative Heat

$$P_{\text{sun}} = \int_0^\infty I_{\text{AM1.5}}(\lambda) \varepsilon(\lambda, \theta_{\text{sun}}) d\lambda \quad (13)$$

$$P_{\text{nonrad}} = h_c (T_{\text{amb}} - T) \quad (14)$$

where $I_{\text{AM1.5}}$ is the ASTM G173-03 solar spectrum ($\sim 1000 \text{ W/m}^2$ total), $\theta_{\text{sun}} = 48.2^\circ$, and $h_c = 6.9 \text{ W/(m}^2 \cdot \text{K)}$.

3.3.5 Performance Metrics

The steady-state equilibrium temperature T_{eq} satisfies $P_{\text{cool}}(T_{\text{eq}}) = 0$, giving the temperature drop $\Delta T = T_{\text{amb}} - T_{\text{eq}}$. The cooling efficiency is $\eta = P_{\text{cool}}/P_{\text{sun,total}}$.

3.4 Problem 3: Multilayer Film Optimization

3.4.1 Multilayer Transfer Matrix Method

For N layers, the total reflection and transmission are obtained by recursive application of Eq. (1). Starting from the bottom layer, the effective reflection coefficient is computed iteratively upward:

$$r_{\text{eff}}^{(j)} = \frac{r_{j,j+1} + r_{\text{eff}}^{(j+1)} e^{2i\beta_{j+1}}}{1 + r_{j,j+1} r_{\text{eff}}^{(j+1)} e^{2i\beta_{j+1}}} \quad (15)$$

3.4.2 Candidate Structures

We evaluate three multilayer configurations:

- **Structure I:** Ag (reflective) / SiO_2 (spacer) / PDMS (emitter)
- **Structure II:** $\text{TiO}_2/\text{SiO}_2$ DBR (solar reflector) / PDMS (emitter)
- **Structure III:** Ag / TiO_2 / SiO_2 / Al_2O_3 (gradient) / PDMS (emitter)

Candidate materials include Ag ($\tilde{n} \approx 0.15 + 3.5i$ visible), SiO_2 ($n \approx 1.46$), TiO_2 ($n \approx 2.6$), Al_2O_3 ($n \approx 1.77$), and PDMS ($n \approx 1.40$ visible).

3.4.3 Optimization Approach

We perform systematic parameter scanning of spacer and emitter layer thicknesses to identify the optimal balance between high 8–13 μm emissivity and low solar absorptance. The design objective is:

$$\max F(d_1, \dots, d_N) = w_1 \cdot \bar{\epsilon}_{8-13} - w_2 \cdot \bar{\alpha}_{\text{solar}} \quad (16)$$

3.5 Problem 4: Comprehensive Design and Feasibility

3.5.1 Cost Model

The per-square-meter cost includes material, processing, and substrate costs:

$$C_{\text{total}} = \sum_{j=1}^N (\rho_j d_j P_j^{\text{mat}} + C_j^{\text{proc}}) + C_{\text{sub}} \quad (17)$$

where ρ_j is density, P_j^{mat} is material price per kg, and C_j^{proc} is processing cost.

3.5.2 Economic Evaluation

Key economic metrics include:

- Annual electricity savings from replaced air conditioning energy
- Payback period (years to recover initial investment)
- 10-year Net Present Value (NPV) with 5% discount rate

3.5.3 Feasibility Assessment

Technology Readiness Level (TRL) is assessed for each fabrication process. Spin-coating (PDMS) and magnetron sputtering (SiO_2 , TiO_2) are at TRL 8–9, while thermal evaporation (Ag) is fully mature at TRL 9.

IV. Solution and Results

4.1 Data Preparation

The PDMS complex refractive index was constructed using a 15-oscillator Drude-Lorentz model calibrated to experimental data from Srinivasan et al. (2016). The AM1.5 solar spectrum follows ASTM G173-03 with total irradiance of 1000 W/m^2 . Atmospheric transmittance in the 8–13 μm window was modeled based on MODTRAN mid-latitude summer conditions, with ozone absorption at 9.6 μm explicitly incorporated.

Figure 1 shows the resulting PDMS complex refractive index. The extinction coefficient k exhibits distinct absorption peaks corresponding to Si–O–Si asymmetric stretching (9.26 μm , $k \approx 0.08$), CH_3 symmetric deformation (7.94 μm , $k \approx 0.06$), and

Si–C rocking vibrations ($12.5 \mu\text{m}$, $k \approx 0.12$). In the visible and near-infrared range ($0.3\text{--}5 \mu\text{m}$), PDMS is essentially transparent with $k < 10^{-5}$.

图 1: PDMS complex refractive index: (top) real part $n(\lambda)$, (bottom) extinction coefficient $k(\lambda)$. The green shaded region indicates the $8\text{--}13 \mu\text{m}$ atmospheric transparency window.

4.2 Problem 1: PDMS Emissivity vs. Wavelength and Thickness

Figure 2 presents the spectral emissivity $\varepsilon(\lambda)$ for PDMS films of varying thicknesses. For thin films ($d < 5 \mu\text{m}$), the emissivity spectrum exhibits pronounced interference fringes due to coherent thin-film interference. As thickness increases, these oscillations dampen and the emissivity converges toward the bulk PDMS absorption profile.

图 2: Spectral emissivity of PDMS thin films at normal incidence. (Top) Thin films ($0.1\text{--}10 \mu\text{m}$) showing interference oscillations. (Bottom) Thick films ($10\text{--}200 \mu\text{m}$) approaching bulk behavior.

Figure 3 shows the average emissivity in the $8\text{--}13 \mu\text{m}$ atmospheric window as a function of film thickness. The emissivity increases rapidly from $\bar{\varepsilon}_{8-13} \approx 0.05$ at $d = 0.5 \mu\text{m}$ to approximately 0.89 at $d = 50 \mu\text{m}$, then gradually saturates toward 0.97 at very large thicknesses ($d > 200 \mu\text{m}$). This saturation behavior reflects the exponential decay of transmitted light: once the optical depth exceeds ~ 3 , the film is effectively opaque in the absorption bands.

图 3: Average $8\text{--}13 \mu\text{m}$ emissivity vs. PDMS thickness. The curve shows rapid initial growth followed by saturation beyond $\sim 100 \mu\text{m}$.

The emissivity heatmap (Figure 4) provides a comprehensive visualization of $\varepsilon(\lambda, d)$, clearly showing the spectral regions of high emissivity coinciding with the PDMS vibrational absorption bands and the atmospheric window.

图 4: Emissivity heatmap $\varepsilon(\lambda, d)$ showing the transition from thin-film interference regime to bulk absorption regime.

4.3 Problem 2: Radiative Cooling Performance

4.3.1 Net Cooling Power Analysis

Figure 5 presents the radiative cooling performance as a function of PDMS film thickness. The net cooling power P_{cool} initially increases rapidly from -8.9 W/m^2 at

$d = 1 \mu\text{m}$ to a peak of 76.5 W/m^2 at $d = 40 \mu\text{m}$, then gradually declines to 47.6 W/m^2 at $d = 300 \mu\text{m}$. The non-monotonic behavior arises from the competition between increasing IR emissivity (favoring thicker films) and increasing solar absorption (disfavoring overly thick films). The optimal thickness of approximately $40\text{--}50 \mu\text{m}$ achieves a favorable balance.

图 5: Radiative cooling performance: (left) power components vs. thickness, (right) temperature drop and cooling efficiency.

4.3.2 Performance Metrics

Table 2 summarizes key performance metrics for representative thicknesses. The $40 \mu\text{m}$ PDMS film achieves the optimal net cooling power of 76.5 W/m^2 with an $8\text{--}13 \mu\text{m}$ emissivity of 0.938 and solar absorptance of only 2.3% .

表 2: Cooling performance for representative PDMS thicknesses

$d (\mu\text{m})$	$\bar{\epsilon}_{8-13}$	$\bar{\alpha}_{\text{solar}}$	$P_{\text{cool}} (\text{W/m}^2)$	$\Delta T (\text{K})$
5	0.485	0.018	31.6	—
20	0.840	0.021	70.7	—
40	0.938	0.023	76.5	—
50	0.952	0.023	76.1	—
100	0.967	0.026	71.7	—

The radar chart (Figure 6) provides a multi-dimensional performance comparison across thicknesses, illustrating the trade-offs between emissivity, solar reflectivity, cooling power, temperature drop, and efficiency.

图 6: Radar chart comparing five performance dimensions for PDMS films at $d = 20, 50$, and $100 \mu\text{m}$.

4.3.3 Key Findings and Recommendations

Based on the evaluation results, we offer the following recommendations:

1. The optimal single-layer PDMS thickness is $40\text{--}50 \mu\text{m}$, balancing high IR emissivity ($\bar{\epsilon}_{8-13} \approx 0.94$) with manageable solar absorption.
2. Beyond $50 \mu\text{m}$, the marginal gain in emissivity is negligible while solar absorption continues to increase, leading to diminishing returns.
3. For practical applications, a reflective substrate (e.g., silver) is essential to minimize solar heating while maintaining structural integrity.

4. The maximum achievable net cooling power of $\sim 77 \text{ W/m}^2$ corresponds to significant annual energy savings when deployed at scale.

4.4 Problem 3: Multilayer Film Optimization

4.4.1 Structure Comparison

Figure 7 compares the spectral performance of three multilayer configurations against a single-layer PDMS baseline. Structure I (Ag/SiO₂/PDMS) with the Ag reflective layer achieves effective solar blocking while the SiO₂ spacer layer provides optical impedance matching, enhancing the effective IR emissivity.

图 7: Spectral comparison of multilayer structures: (left) 6–15 μm IR emissivity, (right) solar-band reflectance.

4.4.2 SiO₂ Spacer Layer Optimization

Figure 8 shows the parametric scan of SiO₂ spacer thickness in the Ag/SiO₂/PDMS structure. The SiO₂ layer has a modest but measurable effect on both IR emissivity and solar absorption through thin-film interference tuning. An optimal SiO₂ thickness of approximately 0.5 μm provides the best balance.

图 8: Effect of SiO₂ spacer thickness on (left) 8–13 μm emissivity and (right) solar absorptance in the Ag/SiO₂/PDMS structure.

4.4.3 PDMS Thickness Optimization with Substrate

Figure 9 demonstrates that when PDMS is supported by an Ag/SiO₂ substrate, the emissivity exhibits a similar saturation trend as the freestanding case. The optimal PDMS thickness remains approximately 40–50 μm , confirming the robustness of this design parameter.

图 9: PDMS thickness optimization on Ag/SiO₂ substrate: (left) IR emissivity, (right) solar absorptance.

4.4.4 Performance Comparison

Figure 10 compares the three multilayer designs against the single-layer baseline. Structure I (Ag/SiO₂/PDMS with optimal thicknesses of 100 nm / 0.5 μm / 50 μm) emerges as the preferred design, offering high IR emissivity ($\bar{\epsilon}_{8-13} = 0.94$) with minimal solar absorption.

图 10: Performance comparison of multilayer structure designs.

4.5 Problem 4: Comprehensive Design and Economic Analysis

4.5.1 Final Design

The optimal comprehensive design is: **Ag (100 nm) / SiO₂ (0.5 μm) / PDMS (50 μm)**, fabricated on a glass substrate using thermal evaporation (Ag), magnetron sputtering (SiO₂), and spin-coating (PDMS).

4.5.2 Cost Analysis

Table 3 presents the detailed cost breakdown per square meter. The total manufacturing cost is estimated at approximately \$40.5/m², with PDMS material cost and SiO₂ processing being the dominant components.

表 3: Cost breakdown for optimal multilayer design

Layer	Thickness	Material (\$/m ²)	Process (\$/m ²)
Ag	100 nm	0.84	10.00
SiO ₂	500 nm	0.22	20.00
PDMS	50 μm	2.43	5.00
Glass substrate	—	2.00	—
Total		3.49	37.00
Total Cost:		\$40.49/m²	

4.5.3 Economic Feasibility

Figure 11 presents the cost-performance Pareto frontier. The optimal design achieves a balance between cooling performance and manufacturing cost. With annual electricity savings of approximately \$8.7/m² (based on COP=3.0 air conditioning replacement), the payback period is estimated at 4–6 years depending on local electricity prices and climate conditions.

图 11: (Left) Performance-cost Pareto frontier across design alternatives. (Right) Economic feasibility assessment showing payback period vs. 10-year NPV.

4.5.4 Technology Comparison

Figure 12 compares radiative cooling with conventional technologies. While the initial investment for radiative cooling is moderate, its zero operating energy consumption provides a decisive advantage over active cooling systems. The technology is particularly suitable for arid regions with clear skies and strong atmospheric window transmittance.

图 12: Comparison of radiative cooling with conventional cooling technologies across normalized metrics.

4.5.5 Feasibility Assessment

All fabrication processes are industrially mature: magnetron sputtering (TRL 9), thermal evaporation (TRL 9), and spin-coating (TRL 8). The primary challenges for large-scale deployment include: (1) ensuring thickness uniformity across large areas, (2) long-term outdoor durability against UV degradation and dust accumulation, and (3) further cost reduction through roll-to-roll manufacturing.

V. Sensitivity Analysis

5.1 Environmental Parameters

We examine the sensitivity of net cooling power to variations in key environmental parameters around baseline values ($T_{\text{amb}} = 300$ K, $h_c = 6.9$ W/(m²·K), clear-sky conditions).

表 4: Sensitivity of P_{cool} to environmental parameters ($d = 50$ μm PDMS)

Parameter	Variation	P_{cool} (W/m ²)	Change
Baseline	—	76.1	—
$T_{\text{amb}} = 295$ K	−5 K	62.4	−18.0%
$T_{\text{amb}} = 305$ K	+5 K	89.8	+18.0%
$h_c = 3.0$ W/(m ² ·K)	Low wind	78.3	+2.9%
$h_c = 12.0$ W/(m ² ·K)	High wind	73.2	−3.8%
Humid atmosphere	Higher H ₂ O	58.7	−22.9%
Arid atmosphere	Lower H ₂ O	88.5	+16.3%

The cooling power shows moderate sensitivity to ambient temperature (~ 3.6 W/m² per K) and strong sensitivity to atmospheric humidity. High water vapor content reduces the atmospheric window transmittance, significantly degrading cooling performance. This highlights the importance of deploying radiative cooling technology in arid and semi-arid climates.

5.2 Fabrication Tolerance

Table 5 examines the impact of $\pm 10\%$ thickness variations on cooling performance. The PDMS thickness is the most critical parameter, with $\pm 10\%$ variation causing approximately ± 2 – 3% change in P_{cool} . The Ag and SiO₂ layers show negligible sensitivity, confirming the robustness of the design against manufacturing tolerances.

表 5: Fabrication tolerance analysis ($\pm 10\%$ thickness variation)

Layer	Thickness Range	P_{cool} Range (W/m ²)
Ag	90–110 nm	76.0–76.2
SiO ₂	450–550 nm	75.8–76.4
PDMS	45–55 μm	74.1–77.8

5.3 Refractive Index Uncertainty

The Drude-Lorentz oscillator parameters have inherent uncertainties from experimental measurements. A $\pm 5\%$ perturbation in the oscillator strengths produces less than 3% variation in the predicted 8–13 μm emissivity, confirming that the model predictions are robust to input parameter uncertainties.

VI. Model Evaluation

6.1 Strengths

1. **Physically rigorous foundation:** The Transfer Matrix Method with Airy summation provides an exact solution to Maxwell's equations for planar multilayer structures, avoiding approximations inherent in effective medium theories. The incorporation of complex refractive indices captures both interference and absorption effects accurately.
2. **Comprehensive cooling model:** The net cooling power calculation integrates blackbody radiation, atmospheric counter-radiation (with angle-dependent emissivity from MODTRAN), solar absorption (AM1.5 spectrum), and non-radiative heat transfer, providing a complete energy balance.
3. **Systematic optimization framework:** The multilayer design methodology progresses logically from single-layer analysis through parameter scanning to comprehensive economic evaluation, enabling informed design decisions.
4. **Computational efficiency:** The Airy summation formulation avoids the numerical instabilities of the conventional characteristic matrix method for absorbing layers while maintaining exactness of the solution.
5. **Robustness:** Sensitivity analysis demonstrates that the optimal design is tolerant to manufacturing variations, supporting practical implementation.

6.2 Weaknesses

1. **Simplified atmospheric model:** The MODTRAN-based atmospheric transmittance is an approximation that does not capture real-time weather variations. A

more sophisticated model incorporating local humidity, aerosol, and cloud data would improve prediction accuracy.

2. **Planar interface assumption:** Surface roughness and interface diffusion, which can affect optical performance in real fabricated devices, are neglected. For spin-coated PDMS on sputtered layers, some interfacial mixing is inevitable.
3. **Lumped non-radiative model:** The single-parameter h_c model for convection does not distinguish between conductive, convective, and radiative parasitic heat paths, each with different temperature and wind-speed dependencies.
4. **Static analysis:** The steady-state assumption does not capture diurnal temperature cycles, transient cloud effects, or seasonal variations, which are important for real-world performance assessment.
5. **Material data limitations:** The Drude-Lorentz oscillator parameters are derived from bulk PDMS measurements and may not perfectly represent thin-film optical constants, which can differ due to molecular orientation and density variations.

6.3 Model Extensions

The modeling framework can be extended to:

1. **Nanostructured surfaces:** Incorporation of grating or photonic crystal structures for enhanced spectral selectivity using rigorous coupled-wave analysis (RCWA).
2. **Dynamic performance:** Transient thermal modeling with hourly meteorological data to predict annual energy savings in specific geographic locations.
3. **Multiphysics coupling:** Integration with building energy simulation tools (e.g., EnergyPlus) for whole-system optimization.
4. **Machine learning acceleration:** Training surrogate models (neural networks) on the TMM physics to enable rapid exploration of vast material and thickness parameter spaces.

VII. Conclusions

This paper has established a comprehensive modeling framework for the design and optimization of PDMS-based radiative cooling technology, addressing four progressively complex sub-problems.

Problem 1 developed a single-layer PDMS emissivity model using the Transfer Matrix Method with Airy summation. The model reveals that PDMS emissivity in the 8–13 μm atmospheric window increases from approximately 0.05 at sub-micron thicknesses to 0.89 at 50 μm , approaching saturation ($\bar{\epsilon}_{8-13} \approx 0.97$) beyond 200 μm . The

spectral features are dominated by Si–O–Si and Si–C vibrational absorption bands at 7.9, 9.3, and 12.5 μm .

Problem 2 evaluated the radiative cooling performance across thicknesses. The net cooling power peaks at 76.5 W/m² for a 40 μm PDMS film, representing the optimal balance between high IR emissivity ($\bar{\epsilon}_{8-13} = 0.938$) and low solar absorptance ($\bar{\alpha}_{\text{solar}} = 0.023$). Beyond 50 μm , diminishing returns set in as increased solar absorption outweighs marginal emissivity gains. Our analysis identifies 40–50 μm as the recommended thickness range for single-layer PDMS radiative coolers.

Problem 3 investigated multilayer structures incorporating reflective and dielectric layers. The Ag (100 nm) / SiO₂ (0.5 μm) / PDMS (50 μm) configuration provides effective solar reflection while maintaining strong IR emission. The SiO₂ spacer layer offers modest but measurable performance enhancement through optical impedance matching. Systematic parameter scanning confirmed the robustness of the optimal thickness parameters.

Problem 4 conducted a comprehensive evaluation encompassing cost analysis, economic feasibility, and technology readiness. The optimal design achieves a manufacturing cost of approximately \$40.5/m² with an estimated payback period of 4–6 years. All required fabrication processes are industrially mature (TRL 8–9). The technology is most suitable for arid and semi-arid regions where the atmospheric window transmittance is highest.

The modeling framework developed herein provides a transferable methodology applicable to other radiative cooling material systems. Future work should focus on experimental validation, nanostructured surface engineering for enhanced spectral selectivity, and integration with building-scale energy simulation for deployment optimization.

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Appendix

Core Physics Functions

Listing 1: Transfer Matrix Method with Airy Summation

```
def transfer_matrix_single_layer(n_layer, d_um, wavelength_um, theta_rad
    ,
                                n_in=1.0, n_out=1.0):
    """
    Single-layer reflectance/transmittance via Airy summation.
    Structure: n_in | n_layer(d) | n_out
    """
    cos_in = cos(theta_rad)
    cos_layer = snell(n_in, n_layer, theta_rad)
    cos_out = snell(n_in, n_out, theta_rad)

    # Phase factor:  $\beta = 2\pi n_{\text{tilde}} d \cos(\theta) / \lambda$ 
    beta = 2.0 * pi * n_layer * d_um / wavelength_um * cos_layer

    R_total, T_total = 0.0, 0.0
    for pol in ['s', 'p']:
        # Fresnel coefficients at interfaces 0->1 and 1->2
        if pol == 's':
            r01 = (n_in*cos_in - n_layer*cos_layer) / \
                (n_in*cos_in + n_layer*cos_layer)
            r12 = (n_layer*cos_layer - n_out*cos_out) / \
                (n_layer*cos_layer + n_out*cos_out)
            t01 = 2.0*n_in*cos_in / (n_in*cos_in + n_layer*cos_layer)
            t12 = 2.0*n_layer*cos_layer / \
                (n_layer*cos_layer + n_out*cos_out)
        else:
            r01 = (n_layer*cos_in - n_in*cos_layer) / \
                (n_layer*cos_in + n_in*cos_layer)
            r12 = (n_out*cos_layer - n_layer*cos_out) / \
                (n_out*cos_layer + n_layer*cos_out)
            t01 = 2.0*n_in*cos_in / (n_layer*cos_in + n_in*cos_layer)
            t12 = 2.0*n_layer*cos_layer / \
                (n_out*cos_layer + n_layer*cos_out)

    exp_2ib = np.exp(2j * beta) # Decays for absorbing media
    denom = 1.0 + r01 * r12 * exp_2ib
```

```

r_total = (r01 + r12 * exp_2ib) / denom
t_total = (t01 * t12 * np.exp(1j * beta)) / denom

R = abs(r_total)**2
# Transmission with impedance correction
if pol == 's':
    T_pol = (n_out.real*cos_out.real) / \
            (n_in.real*cos_in.real) * abs(t_total)**2
else:
    T_pol = (n_out.real/cos_out.real) / \
            (n_in.real/cos_in.real) * abs(t_total)**2

R_total += R; T_total += T_pol

return np.clip(R_total/2, 0, 1), np.clip(T_total/2, 0, 1)

```

Listing 2: Planck Blackbody Spectral Radiance

```

def planck_spectral_radiance(wavelength_um, T):
    """
    I_BB(lambda,T) = 2hc^2/lambda^5 * 1/(exp(hc/(lambda*k_B*T)) - 1)
    Returns: W/(m^2.um.sr)
    """
    wl_m = wavelength_um * 1e-6
    exponent = h * c / (wl_m * kB * T)
    with np.errstate(over='ignore'):
        radiance_per_m = 2.0*h*c**2 / (wl_m**5) / (np.exp(exponent)-1.0)
        radiance = radiance_per_m * 1e-6 # Convert to per-micrometer
    radiance = np.where(np.isfinite(radiance), radiance, 0.0)
    return radiance

```

Listing 3: Net Cooling Power Calculation

```

def compute_net_cooling_power(d_um, wavelengths_um, n_arr, k_arr,
                              T, T_amb, h_c=6.9, n_sub=1.0+0j,
                              theta_sun=np.radians(48.2), n_angles=12):
    """P_cool = P_rad - P_atm - P_sun - P_nonrad"""
    # Setup Gauss-Legendre angular quadrature
    x, w = leggauss(n_angles)
    theta_arr = (x + 1.0) * pi / 4.0
    weights = w * pi / 4.0

    # P_rad: Hemispherical integration of Planck emission
    P_rad = 0.0

```

```

for theta, wt in zip(theta_arr, weights):
    emis = compute_emissivity_spectrum(d_um, wavelengths_um,
                                       theta, n_arr, k_arr, n_sub)
    I_bb = planck_spectral_radiance(wavelengths_um, T)
    P_rad += 2*pi * sin(theta)*cos(theta) * wt * \
        simpson(I_bb * emis, wavelengths_um)

# P_atm: Atmospheric counter-radiation
t_atm = build_atmospheric_transmittance(wavelengths_um)
P_atm = 0.0
for theta, wt in zip(theta_arr, weights):
    emis = compute_emissivity_spectrum(d_um, wavelengths_um,
                                       theta, n_arr, k_arr, n_sub)
    eps_atm = 1.0 - t_atm ** (1.0/max(cos(theta), 1e-6))
    I_bb_atm = planck_spectral_radiance(wavelengths_um, T_amb)
    P_atm += 2*pi * sin(theta)*cos(theta) * wt * \
        simpson(I_bb_atm * emis * eps_atm, wavelengths_um)

# P_sun: Solar absorption (AM1.5)
emis_sun = compute_emissivity_spectrum(d_um, wavelengths_um,
                                       theta_sun, n_arr, k_arr,
n_sub)
I_sun = build_am15_spectrum(wavelengths_um)
P_sun = simpson(I_sun * emis_sun, wavelengths_um)

# P_nonrad: Convection/conduction
P_nonrad = h_c * (T_amb - T)

P_cool = P_rad - P_atm - P_sun - P_nonrad
return {'P_rad': P_rad, 'P_atm': P_atm, 'P_sun': P_sun,
        'P_nonrad': P_nonrad, 'P_cool': P_cool}

```

Full code (problem1.py – problem4.py, utils.py) is available in the code/ directory. The implementation uses NumPy for vectorized computation, SciPy for numerical integration (Gauss-Legendre quadrature and Simpson’s rule), and Matplotlib for visualization. All figures are generated as PDF vector graphics suitable for publication.